# ITM UNIVERSITY, GWALIOR

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## MACHINE LEARNING RESEARCH PBL

# QUBITWATCH: PREDICTING TEMPORAL NOISE DRIFT IN QUANTUM COMPUTING SYSTEMS

### PART 2: STRUCTURED LITERATURE SURVEY AND RESEARCH GAP REFINEMENT

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**RESEARCH PBL PART 2 | STRUCTURED LITERATURE SURVEY**

Focus: next-recorded IBM Quantum readout-error prediction

20 marks | Submission: 5 November 2026 | Due: 11:59 PM IST

01 | CONTEXT AND REVIEW PROTOCOL

## 1. Project context and scope

QubitWatch is framed here as a classical machine-learning study of quantum-device telemetry. The target is the next recorded readout-error value for the same backend and physical qubit. Predicting this reported quantity is different from predicting a quantum state, correcting a measured distribution, or identifying the microscopic cause of noise. A forecast may support awareness of changing device quality, but its operational value must be tested separately from its statistical accuracy.

The confirmed project context is IBM Quantum readout-error forecasting. Earlier discussion names Random Forest, XGBoost and LightGBM as project model families. The poster, notebook and dataset were not retrievable in the referenced chat or this workspace during preparation. Consequently, their schema, date range, device coverage, sample counts, train/test split and reported scores are not asserted. Model comparisons below are refined objectives, not a reconstruction of unverified notebook experiments.

## 2. Structured search and selection

A targeted, structured search was conducted on 5 November 2026 using scholarly title searches, arXiv records, publisher pages and author or institutional repositories. Query families included "quantum calibration drift forecasting readout error", "detecting and tracking drift quantum processors", "predictive estimation qubit dynamics", "measurement error mitigation" and "time series predictor evaluation". Publication identity and findings were checked against primary-source abstracts, publisher descriptions and accessible article text. This is a focused literature survey, not an exhaustive systematic review or a PRISMA study; no unrecoverable search-hit totals are invented.

**Inclusion:** identifiable scholarly papers with a stable DOI or repository URL, and a direct contribution to drift physics, tracking/control, readout characterization, prediction, hardware variability or forecast evaluation. **Exclusion:** unverifiable citations, generic commentary, unrelated quantum machine learning, and resources without sufficient evidence for the attributed finding. Published versions are cited when verified; [15] is explicitly a preprint. IBM documentation [16] supplies terminology and is not counted among the 15 papers.

## 3. Extraction and attribution rules

For every selected paper, the comparison records its problem, method, actual reported finding and boundary relevant to QubitWatch. "R" denotes a restriction or caveat explicitly supported by the source; "S" denotes this survey's scope inference from the paper's task, platform or data. An S statement is not presented as an author-admitted defect. Numerical results are retained only where directly verifiable, and are never transferred to QubitWatch.

> **Reading the evidence:** evidence that noise changes over time does not establish that its next recorded value is predictable. Likewise, successful mitigation of present measurements does not establish future readout-error forecast skill.

02 | THEMATIC LITERATURE SYNTHESIS

## 4. What the literature establishes

**Temporal variability has a physical basis.** Klimov et al. [1] link large relaxation-time fluctuations to two-level-system defects. Burnett et al. [2] use multi-day observations to show that coherence parameters vary and that relaxation fluctuations can be local. These studies motivate repeated observation and qubit-specific analysis. Their relaxation and dephasing results cannot be treated as evidence that an IBM readout-error series has the same dynamics.

**Tracking and prediction already exist.** Proctor et al. [3] resolve time-varying errors through spectral analysis and demonstrate drift suppression in trapped-ion experiments. Gupta and Biercuk [4] explicitly study forward prediction from historical projective measurements and show that model choice affects the prediction horizon. Kelly et al. [5] demonstrate in-situ tuning during repetitive error detection. Together they defeat a claim that drift is universally ignored, while leaving a distinct question about passive prediction from archived IBM property records.

**Readout characterization is richer than one scalar.** Detector tomography [6], tensor-product and correlated models [7], and cross-talk characterization [8] describe measurement errors at different levels. Nation et al. [9] address mitigation scalability, while Nachman et al. [10] examine statistical instability in unfolding. Kim et al. [11] use a neural network to correct measured output distributions. These methods operate on distributions, assignment models or calibration experiments; a single next-recorded error forecast cannot reproduce their information or replace their calibration procedures.

**Device quality matters for applications.** Murali et al. [12] and Tannu and Qureshi [13] show that hardware-aware mapping can improve program reliability in their tested settings. This motivates monitoring device quality, but a regression score alone cannot establish better compilation or circuit success. Such benefits would require a separate downstream comparison, beyond this literature-review stage.

**Forecast evaluation is part of the research problem.** Bergmeir and Benitez [14] examine time-series evaluation and advocate blocked cross-validation in their study. Their findings do not imply that every use of random cross-validation is invalid. For QubitWatch, forward evaluation is a design choice matched to deployment: the model must predict a later record using information actually available earlier. A recent preprint [15] also uses past observation windows to forecast open-system dynamics, reinforcing that temporal ML itself is not a novel contribution.

## 5. Consequences for QubitWatch

The project should prioritize target validity, chronology and baseline comparison over model complexity. A high score can arise from stable qubit identities, duplicate property measurements, or a slowly varying target. These alternatives must be checked before interpreting performance as learned temporal drift. The intended contribution is evidence about forecastability on a clearly bounded archive, including a negative result if persistence remains competitive.

> **Three distinct tasks:** characterization estimates present noise; mitigation corrects affected observations; forecasting estimates a later quantity. QubitWatch addresses the third task. The reviewed papers inform it without becoming interchangeable benchmarks.

03 | EVIDENCE COMPARISON (1/3)

## 6.1 Comparison of selected papers

R = source-reported caveat; S = survey inference about scope. Each numbered citation links the row to its full reference on pages 9-10.

| Author / year | Problem | Method | Actual finding | Limitations / boundary |
|---|---|---|---|---|
| 1. Klimov et al. (2018) [1] | Why superconducting-qubit relaxation times fluctuate. | Spectral and temporal probing of individual two-level-system defects. | Direct evidence links the largest relaxation fluctuations to these defects. | S: Relaxation-time physics, not next-recorded IBM readout-error regression. Use as motivation, not predictive validation. |
| 2. Burnett et al. (2019) [2] | Reproducibility of coherence parameters over time. | Multi-day measurements and statistical analysis of relaxation, dephasing and frequency. | Both coherence quantities fluctuate; relaxation fluctuations are local and linked to near-resonant defects. | S: Measured coherence dynamics differ from a scalar readout metric. Locality motivates per-qubit analysis, not a universal shared drift law. |
| 3. Proctor et al. (2020) [3] | Static characterization can miss time-dependent errors. | Spectral tests, time-resolved model selection and parameter estimation. | Simulations and trapped-ion experiments resolve instability; subsequent control suppresses it. | R: Sensitivity depends on sampling rate and observation duration. S: Detection/tracking is not held-out IBM telemetry forecasting. |
| 4. Gupta & Biercuk (2018) [4] | Forward prediction of a dephasing qubit from past observations. | Autoregressive and Fourier Kalman filters; Gaussian-process alternatives. | Autoregressive filtering outperforms Fourier variants; tested GPR kernels are generally unsuitable for forward prediction. | R: Performance depends on filter optimization and prediction horizon. S: Single-qubit dephasing differs from readout-error archive regression. |
| 5. Kelly et al. (2016) [5] | Control-parameter drift during repetitive error detection. | Active Detection Event Parameter Tuning (ADEPT) with parallel tuning. | A nine-qubit experiment optimizes gates and compensates inserted single-qubit and simultaneous independent frequency drifts. | S: Requires error-detection feedback and controllable parameters; QubitWatch uses passive records and cannot claim equivalent drift compensation. |

These five studies establish variability, temporal inference and feedback control as prior art. Their platforms and target quantities differ. The project must therefore describe its novelty at the level of archive, target and evaluation rather than claim to introduce quantum-noise forecasting.

03 | EVIDENCE COMPARISON (2/3)

## 6.2 Comparison of selected papers

R = source-reported caveat; S = survey inference about scope. Each numbered citation links the row to its full reference on pages 9-10.

| Author / year | Problem | Method | Actual finding | Limitations / boundary |
|---|---|---|---|---|
| 6. Maciejewski et al. (2020) [6] | Readout noise biases measurement statistics. | Quantum detector tomography followed by classical correction. | Classical readout noise dominates on tested IBM/Rigetti devices; correction improves several tasks on an IBM five-qubit device. | R: Exact correction requires invertible classical noise; coherent errors and finite statistics affect performance. S: Correction is not temporal forecasting. |
| 7. Bravyi et al. (2021) [7] | Mitigating multi-qubit measurement errors. | Tensor-product and correlated Markovian noise models; inverse-noise post-processing. | Demonstrations include graph-state stabilizers up to 12 qubits and entangled 20-qubit states on IBM devices. | R: Overhead is exponential in maximum error rate times qubit count. S: Calibration-derived correction does not evaluate next-record error prediction. |
| 8. Maciejewski et al. (2021) [8] | Correlated measurement cross-talk and its effect on QAOA. | Local correlated-noise model; Diagonal Detector Overlapping Tomography; marginal mitigation. | Average error reduction exceeds 22-fold on 15 IBM qubits and 5.5-fold on 23 Rigetti qubits in reported tests. | R: The model uses restricted locality and bounded residual error. S: These mitigation factors are not forecast metrics; a scalar target omits joint correlations. |
| 9. Nation et al. (2021) [9] | Memory and computation cost of measurement-error mitigation. | Reduced bit-string subspace and matrix-free preconditioned iterative solution. | Demonstrates mitigation in seconds with orders-of-magnitude less memory than direct factorization. | S: Addresses scalable correction using a noise description, not learning future calibration values. Computational efficiency does not demonstrate robustness to future drift. |
| 10. Nachman et al. (2020) [10] | Unstable inversion of readout response matrices. | Comparison of unfolding methods, including iterative Bayesian unfolding. | Iterative Bayesian unfolding avoids pathologies of common inversion and least-squares approaches in studied examples. | R: Inversion can amplify statistical fluctuations; low-count regions remain uncertain. S: Correcting counts with a response model is distinct from forecasting model change. |

The five studies show why readout errors require careful characterization and why joint error information can matter. Forecasting one reported marginal error value is a narrower task. Mitigation improvements cannot be placed in the same numerical ranking as MAE or RMSE.

03 | EVIDENCE COMPARISON (3/3)

## 6.3 Comparison of selected papers

R = source-reported caveat; S = survey inference about scope. Each numbered citation links the row to its full reference on pages 9-10.

| Author / year | Problem | Method | Actual finding | Limitations / boundary |
|---|---|---|---|---|
| 11. Kim et al. (2022) [11] | Reducing readout errors beyond linear correction. | Artificial neural network trained on measurement results from simple single-qubit-gate circuits. | Reports an advantage over existing methods in mitigation experiments on IBM five-qubit devices. | S: The learned mapping corrects measured distributions; it is not a next-recorded scalar forecast. Device-tested mitigation does not establish temporal generalization. |
| 12. Murali et al. (2019) [12] | Mapping programs onto variable-quality NISQ hardware. | Compiler mappings using calibration data, routing choices and reliability estimates. | Real-system tests report 2.9-fold average and up to 18-fold program-success improvement over the compared Qiskit compiler. | S: Hardware-aware allocation uses quality information; it does not validate future readout forecasts. Reported gains depend on the tested compiler and workloads. |
| 13. Tannu & Qureshi (2019) [13] | Nonuniform qubit/link quality in allocation and movement. | Variability-aware qubit allocation and movement policies. | IBM-Q5 evaluation reports reliability improvement up to 1.9-fold. | S: Spatial variability-aware mapping differs from longitudinal prediction. The reported application gain cannot be inferred from QubitWatch regression accuracy. |
| 14. Bergmeir & Benitez (2012) [14] | How to evaluate time-series predictors and select models. | Empirical comparison of evaluation and cross-validation procedures. | Cross-validation gives robust model selection in the study; authors recommend a blocked form for time series. | R: Their study does not find practical effects of all theoretical flaws. S: It is methodological support, not a quantum benchmark or universal ban on random splits. |
| 15. Abu-Nada & Banerjee (2025) [15] | Forecasting open-system observables and detecting memory. | Feedforward network with short past windows of ancilla data; revival-based score. | For studied amplitude-damping and telegraph-noise channels, forecasts reproduce dynamics and flag memory; telegraph noise has a larger revival score. | S: Model-channel study concerns observables, not archived IBM readout errors. Preprint status: no peer-reviewed publication is asserted here. |

These studies connect learning, device-aware decisions and evaluation. They motivate model comparison and deployment-matched testing, but none of the extracted findings establishes QubitWatch performance. The recent preprint is used as related work with its publication status stated.

04 | RESEARCH GAP REFINEMENT

## 7. From broad motivation to a defensible gap

**Initial formulation:** "Predicting temporal noise drift in quantum computing systems." This is a useful project theme, but it combines several physical mechanisms, platforms and targets. It is too broad for a claim of novelty. Predictive dephasing estimation already appears in [4], drift detection and tracking in [3], active compensation in [5], and observable forecasting in [15].

**Evidence-bounded gap:** the selected corpus does not establish whether inexpensive, passive models trained on historical IBM backend/qubit property records predict the next recorded scalar readout error better than persistence under forward-in-time evaluation. This is a gap in the evidence assembled for this project, not proof that no such study exists anywhere. The literature motivates the test; the archive and future evaluation determine whether the opportunity is empirically useful.

## 7.1 Four boundaries that make the gap testable

**Target boundary.** Forecast the same device/qubit's next recorded readout error, not T1, T2, a full assignment matrix or circuit fidelity. Studies [1-2] motivate temporal observation; [6-11] show why a scalar readout metric only represents part of measurement noise.

**Information boundary.** Use only records and metadata available at forecast issuance. A property measured later but stored in an earlier-looking download cannot be an input. IBM distinguishes monitoring, calibration and benchmarking, with individual property timestamps [16]. Thus a row's download time must not automatically be interpreted as the readout measurement time.

**Evaluation boundary.** Compare against the last known value on the same chronological test records. Identity effects, repeated snapshots and train/test overlap can make a model appear useful without learning temporal change. This follows from the forecasting setup and is informed by the evaluation discussion in [14].

**Generalization boundary.** Start with future observations on observed qubits/backends. Prediction for unseen hardware is a separate, harder claim and requires an explicit group-held-out test. Cross-talk evidence [8] also prevents interpreting a good scalar forecast as a complete correlated-noise model.

## 7.2 Refined research question

> **RQ:** For each backend-qubit series in the QubitWatch archive, can historical readout-error values and available, timestamp-valid device metadata reduce next-recorded readout-error MAE relative to persistence on later, unseen records, and how stable is any gain across qubits, time blocks and elapsed observation gaps?

**Falsifiable expectation:** lagged temporal information may add value beyond identity and the latest error value. This expectation is rejected if improvement disappears under deduplication, chronological evaluation or per-group reporting. A null result would still define the practical limits of the archive.

05 | OBJECTIVES AND EVALUATION REQUIREMENTS

## 8. Refined objectives

**O1 - Establish the data contract.** Audit backend identity, physical-qubit identity, readout-error definition, units, property timestamps and duplicate measurements. Document available coverage and missingness. This is a prerequisite; no undocumented record count or observation cadence is assumed.

**O2 - Construct causal next-record pairs.** Sort unique measurements within each backend-qubit series. For record i, define y = e(i+1) and inputs from records up to i only. Retain observed elapsed gaps for analysis. The time to the next record may be unknown at issuance, so it is not an input unless a collection schedule makes it known in advance.

**O3 - Assess incremental forecast skill.** Compare persistence, a past-only rolling mean and a regularized linear lag model with the project's reported tree-model families (Random Forest, XGBoost and LightGBM), subject to notebook verification. Keep tuning inside earlier training/validation periods and evaluate all models on identical eligible test pairs.

**O4 - Explain where skill holds.** Compare latest-error-only, lag-history and timestamp-valid metadata feature sets. Report results by qubit/backend, chronological block and observed record gap. Interpret feature importance as predictive association, not proof of physical causes.

## 9. Minimum evaluation contract for subsequent work

**Chronology:** reserve a final contiguous time block before tuning and use expanding-window validation inside the earlier period. Group same-time device records together. Include a training pair only if its target measurement was already available before the validation/test issuance boundary. Fit imputation, encoding and scaling on training data only. Do not interpolate with future values or shuffle time records for the primary evaluation.

**Metrics:** primary MAE = mean |y - prediction|; secondary RMSE emphasizes larger misses. Report error as a probability and, if useful, percentage points. Define persistence skill = 1 - MAE(model)/MAE(persistence); positive values indicate improvement, and zero-denominator cases are reported separately. R-squared is secondary and may be unstable for near-constant targets. Include both overall error and equally weighted per-series summaries.

**Uncertainty and robustness:** report paired model-minus-persistence errors on matched records. If sample coverage permits, use time-block resampling for uncertainty, respecting temporal dependence. Repeat with a later test period and check duplicate/long-gap sensitivity. Missing target values are not fabricated; insufficient histories and exclusions must be counted in the later empirical report.

## 10. Conclusion and Part 2 outcome

The review narrows QubitWatch to an auditable forecastability study. Its contribution would be a validated target construction and a transparent test of incremental skill over simple baselines. The literature supports the motivation and cautions against broad novelty claims. Experimental scores, model superiority, causal explanations and calibration benefits remain unproven; they belong to subsequent work after the original project artifacts are verified. This submission completes the literature comparison, gap refinement and objective definition required for Part 2.

06 | REFERENCES (1/2)

## 11. References

Scholarly corpus: [1]-[15]. Publication years refer to the cited journal/conference version where verified; linked repository versions may have earlier dates. DOI and source URLs are clickable.

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06 | REFERENCES (2/2)

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https://arxiv.org/abs/1910.01969

**[11] J. Kim, B. Oh, Y. Chong, E. Hwang and D. K. Park (2022).** Quantum readout error mitigation via deep learning. *New Journal of Physics 24, 073009.*
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**Evidence note:** this survey uses paraphrased source findings and explicitly identified scope inferences. It makes no claim to reproduce the cited experiments or to verify unavailable QubitWatch artifacts. Published mitigation and application improvements remain attributed to their original studies.